Measuring Contextualized Alignment in Situated LLM Decision-Making

Completed Research Paper

Introduction

Imagine your firm is overwhelmed with applicants to a job posting. To automatically highlight candidates that fit the position best, you develop an artificial intelligence (AI) system that pre-screens applicants. At the core of the system you rely on a large language model (LLM) to take a decision based on a variety of unstructured data in the application documents. To make sure that the model decides in line with your intention and values, you provide some of your requirements, preferences, and the firm's culture explicitly in the prompt. In other words, you make some of the implicit organizational knowledge explicit to better *align* the LLM decision to your specific context. But how do you know that it works? Can you be certain that the LLM has sufficient knowledge to decide adequately for your setting?

Many organizations are currently exploring similar use-cases that involve LLM decision-making. But – although the capabilities of LLMs are rapidly increasing – real-world returns remain below expectations (Challapally et al., 2025). In controlled environments the models outperform most humans at most tasks, as evident by the fast saturation of existing benchmarks (Akhtar et al., 2026; Chang et al., 2024). In the real world, they often underperform "due to brittle workflows, lack of contextual learning, and misalignment with day-to-day operations" (Challapally et al., 2025, p. 3). Nevertheless, the use-cases for LLM decision-making are broad and promising: They provide medical diagnoses (Gaber et al., 2025), choose actions on the web (Yang et al., 2024), classify documents (Sun et al., 2023), assess credit risk (Golec & Alabduljalil, 2026), perform legal judgment (Shui et al., 2023), and select tools in agentic workflows (Wang et al., 2024). In these situations the adequate decision often depends on the environment, i.e. the sociotechnical system that the LLM is deployed in (Dolata et al., 2022; Johnson & Verdicchio, 2025; Kastner, 2025; Pakarinen & Huising, 2023; Selbst et al., 2019). The contextual and situational knowledge relevant for this is by default missing in AI models (Arzberger et al., 2025; Lebovitz et al., 2021), leading to the previously described failure modes.

However, unlike prior machine learning systems, which required retraining or manual feature engineering to incorporate domain knowledge, LLMs can learn from natural language instructions and examples provided at inference time, a capability known as in-context learning (Brown et al., 2020). This property makes a new class of alignment interventions feasible: By articulating organizational knowledge in text and passing it to the model through the prompt, developers can in principle move LLM decision-making from generic to situated without modifying the model's parameters (Kemp, 2023). Yet it remains unclear how to measure whether an LLM's decisions are actually aligned with the organizational context it operates in.

Information systems (IS) research is uniquely positioned to address this. It has long emphasized the interdependence of technical and social subsystems in an organization (Chatterjee et al., 2021). In this view, developing technical components of an organization needs to simultaneously consider the social components to improve the interaction between them (Lee, 2010; Li, 2024). With LLMs pushing the frontier of technical subsystems towards higher agency and less supervision (Baird & Maruping, 2021; Berente et al., 2021; Zhang et al., 2021), the alignment with the social subsystem becomes both more consequential and harder to verify. This sociotechnical perspective thus foregrounds an alignment question that reaches beyond how AI alignment is traditionally understood in computer science. Instead of asking "Is the AI system aligned to human intentions and values?" (Gabriel, 2020; Ji et al., 2025; Russell, 2019), we can thus ask "Is the AI subsystem aligned to the intentions and values of the sociotechnical system it is deployed in?".

In this paper, we propose the *contextualized alignment lens model* (CALM) – a method to measure the degree of this alignment to the application context. The conceptual background of the method builds on the philosophical limits of language and organizational knowledge creation theory. It then extends the Brunswik lens model (Brunswik, 1956) – a psychological model of decision-making under uncertainty – to analyze cue weighting policies to provide a quantitative estimate of how well the LLM decision-making process matches the one of the organization's past decisions. This allows to identify where misalignment occurs and crucially, gives a more detailed insight into what interventions might be helpful to improve alignment, based on errors in cue weighting and hypothesized knowledge intervention effects. Additionally, it provides an estimate into how much of the decision-making is influenced by tacit or implicit knowledge, which gives implications for the automation potential of a given decision.

We demonstrate CALM on 30,000 LLM decisions across ten models and three prompting conditions on European Court of Human Rights (ECtHR) decisions on Article 6. The demonstration delivers evidence on CALM's affordances. For intervention design, CALM guides the design, and measures the size as well as the direction of effects of two prompt-based knowledge externalization interventions at the cue level. This reveals that externalizing some of the organization's implicit knowledge embedded in the decision policy moves cue-utilization closer to the organization's in most cases, with the largest effects on the models with a largely misaligned starting point. We also observe that, even if the knowledge externalization interventions improve alignment, a gap remains, which provides an upper bound for the effect of implicit and tacit knowledge in a given decision. This has implications for the automation potential. Two further interesting findings were surfaced by the framework: Process and outcome alignment correlate strongly across model–condition combinations, and baseline alignment varies widely across models in ways that general benchmarks and pricing tier do not predict.

This paper makes two contributions. Conceptually, we propose that a knowledge gap between organizational and LLM decision-makers motivates the construct of contextualized alignment, which emphasizes the importance of aligning the available knowledge structure for a situated LLM decision to allow it to reach the same outcome with the same process. Simultaneously, we recognize that in many cases the knowledge transfer to LLMs is fundamentally limited due to the tacit nature of some of the relevant organizational knowledge. Methodologically, we provide a tool to measure this degree of alignment by comparing decision policies. The remainder of the paper develops the conceptual background, the method and its diagnostic information, the demonstration design and results, and a discussion of implications, limitations, and future research.

Conceptual Background

The Knowledge Gap in LLM Decision-Making

Context is often treated as a technical variable that developers inject into prompts, retrieval pipelines, or tool descriptions. Work across philosophy and organization theory suggests a stronger reading. Context determines which distinctions are salient, which background assumptions are activated, and which standards of adequacy a decision must satisfy (Weber & Glynn, 2006). A context-blind evaluation therefore omits both the information a decision needs and the frame within which correctness is defined. For a LLM, the context is usually conveyed through text.[1] What the model knows for a decision is therefore whatever the textual input carries about the organizational context in which the decision arises. Two properties of this channel explain why it always carries only a partial representation of organizational knowledge.

The first concerns *what can be articulated*: some organizational knowledge is ineffable. Polanyi first described this tacit dimension, in which skilled practitioners know more than they can tell (Polanyi, 1962, 1966; Tsoukas, 2002). Wittgenstein (1953) further argued that linguistic meaning depends on the "language-game" in which expressions are used, so language alone cannot transfer the full meaning to someone unfamiliar with the underlying practice. For language models this means that a residual gap between organizational knowledge and explicit input is unavoidable whenever tacit or ineffable practice is relevant, because language cannot transmit what language cannot express. Recent management and IS work develops this point further: Pakarinen and Huising (2023) argue that professional expertise is relationally constituted, meaning that it is enacted through ongoing interactions among professionals, clients, and instruments, rather than residing in any single mind or document that could be transferred to an AI system. Lebovitz et al. (2021) show in a field study of machine-learning tools at a major hospital that

[1] We focus on text as the majority of context in organizational applications (codified knowledge, input prompt, instructions etc.) is provided through this channel. However, we acknowledge that many modern LLMs allow for multimodal input and recognize this as limitation. As soon as a significant part of the context is provided through photo/video/sensory input, the frontier of what *can* be conveyed might shift.

even when "ground truth" labels to train the AI are produced by qualified experts, those labels capture experts' codified know-what but miss the rich know-how that experts actually draw on in practice. Together, these studies extend the philosophical argument with empirical evidence: the gap between codified expert knowledge and practiced expertise constrains what an LLM can be told through its prompt and equally constrains what it can learn from training on organizational know-what.

The second concerns *what is articulated*: even where organizational knowledge could in principle be expressed, firms do not fully externalize, codify, or disseminate it. Organizational knowledge creation theory treats externalization as a central process through which implicit know-how becomes communicable and shareable (Nonaka, 1994). Yet the successful transfer of knowledge depends on its codifiability and teachability, is often impeded by internal 'stickiness,' and remains constrained by the extent to which knowledge is embedded in interactions among people, tools, and tasks (Argote & Ingram, 2000; Szulanski, 1996; Zander & Kogut, 1995). What reaches the model through prompts, tool descriptions, and retrieved documents is therefore a partial selection from a larger body of documented and undocumented organizational practice.

Taken together, these two properties define the epistemic position of an LLM acting as a decision unit. The model's effective knowledge for a given decision depends not only on its encoded world knowledge but also on the provided context. The textual input (prompt, retrieved documents, tool descriptions, conversational history etc.) functions as epistemic context, shaping what the model can justifiably infer at that moment. An LLM given thin or poorly targeted context therefore has less actionable knowledge for a decision than the same model given richer and better-structured context, even if its underlying capacities are unchanged. We refer to the combined effect caused by limits of fundamental expressibility and actual articulation as the *knowledge gap*. The gap operates twice in the decision process: (1) It constrains what reaches the model as input, and (2) it shapes which elements of the available context the model treats as relevant to the decision at hand. Ecological rationality frames the second operation as the fit between an agent's decision-making resources and the structure of the decision environment (Luan et al., 2019). Both effects become visible with our method.

From AI Alignment to Contextualized Alignment

AI alignment originated as a research agenda concerned with whether AI systems pursue objectives that match human intentions and values (Gabriel, 2020; Ji et al., 2025; Russell, 2019). The dominant technical instantiation is alignment through preference-based training, which tunes LLMs so that their outputs conform to aggregated human judgments over model behavior (Christiano et al., 2017; Ouyang et al., 2022). Although weaknesses of this method exist (see for example Carlsmith, 2023; Greenblatt et al., 2024; Krakovna et al., 2020; Shane et al., 2026), it has worked well to improve the general usefulness of generative models. A more fundamental limitation, however, follows from the aggregation step itself: Preferences elicited from diverse raters are collapsed into a single reward signal, which suppresses the distribution of values in the underlying population (Sorensen et al., 2024) and, by the same mechanism, fails to capture the situated norms, conventions, and practices that govern any specific organization's decisions (Arzberger et al., 2025; Lebovitz et al., 2021; Pakarinen & Huising, 2023). Pluralistic alignment is one response to this aggregation problem, motivated by population-level diversity (Sorensen et al., 2024). *Contextualized alignment*, which we highlight here, is a parallel response motivated by organization-level specificity.

A discrete LLM decision is contextually aligned when two conditions hold. First, the decision is normatively correct for the organizational context in which it is embedded. This requirement is qualitatively different from general helpfulness or harmlessness, because normative correctness is context-dependent: Even a value as widely endorsed as fairness is constituted within the specific sociotechnical system in which a decision is made and cannot be settled by a context-free definition (Dolata et al., 2022). Second, the decision is reached through a process that an informed organizational decision-maker would endorse. This requirement is qualitatively different from outcome agreement, because two different processes can produce the same output (Geirhos et al., 2020). We refer to the first condition as *outcome alignment* and to the second as *process alignment*. We operationalize the second condition through the cue-utilization policy revealed by past organizational decisions, which we treat as a reasonable approximation of an informed decision-maker's policy in stable institutional settings. This choice rests on the premise that past decisions encode the organizational knowledge that constitutes a firm's competitive advantage (Grant, 1996): the routines, judgments, and accumulated know-how that are not fully codifiable but become visible in the pattern of decisions the organization has actually made. We acknowledge that past decisions can also encode biases that the organization itself would reject on reflection and return to this caveat in the limitations.

The distinction between outcome and process matters for measuring alignment. Outcome alignment can be assessed by comparing LLM decisions to organizational ground truth where such ground truth exists. Process alignment requires access to the decision-relevant cues the model used and the weights it placed on them, because a model may reach the correct output by relying on cues the organization would reject. This phenomenon is well-documented in the broader machine-learning literature on shortcut learning, where decision rules perform well on benchmark distributions but rest on features that have no causal connection to the underlying task (Geirhos et al., 2020). On top of that, in LLMs, the stated reasoning process in a chain-of-thought can also misrepresent the actual decision process (Lanham et al., 2023; Turpin et al., 2023). A deployment that is outcome-aligned but process-misaligned is therefore fragile and poses a security risk. Process alignment is particularly consequential for decisions on judgmental tasks (Laughlin & Ellis, 1986), where the output is not immediately verifiable and organizational standards thus depend on decision process adequacy rather than accuracy alone. Measuring process alignment therefore requires a structure for representing both the organization's and the LLM's decision policy on equal terms, which is exactly what the Brunswik lens model delivers.

The Brunswik Lens Model as a Bridge

The Brunswik lens model offers a tool for analyzing how decision-makers integrate information under uncertainty (Brunswik, 1956; Karelaia & Hogarth, 2008). In Brunswik's probabilistic functionalism, a judge cannot observe the distal criterion of a decision directly and must infer it from observable cues, each of which carries only partial validity (Dhami & Mumpower, 2018). The judge's decision policy is the pattern of weights placed on these cues, recoverable empirically by regressing observed decisions on cue values (Hammond et al., 1964). Hammond's extension of this idea into policy capturing and judgment analysis established the lens model as a tool for studying expert judgment in clinical, educational, and managerial domains (Karelaia & Hogarth, 2008; Karren & Barringer, 2002).

Three properties make this tool a natural fit for our problem. First, the structure of probabilistic functionalism, with an uncertain distal criterion mediated by observable cues, matches the structure of organizational decisions whose adequate outcome is not directly accessible and whose justification rests on context-dependent features of the case. Second, the lens is symmetric: the same cue regression can be applied on the environmental side and on the judge side, which yields a single comparable representation of policy on each side (Hammond et al., 1964). Smith et al. (2003) exploit this symmetry to compare clinical guidelines for the treatment of depression with the prescribing policies of individual general practitioners, finding systematic differences in cue weighting that output agreement alone would not have revealed. We adopt the same comparative move in CALM, with the organizational policy taking the place of the guidelines and the contextualized LLM taking the place of the individual judge. Third, the lens model treats the cue-utilization policy as a behavioral inference rather than a description of internal cognition, which makes it as applicable to systems whose internal mechanisms are opaque as it is to human judges.

The Contextualized Alignment Lens Model (CALM)

The Brunswik lens supplies a structure for comparing decision policies that operate on the same observable cues. We use this structure to compare the decision policy of a contextualized LLM with the organizational decision policy that it should reproduce to be contextually aligned. The organizational policy is the alignment target (analog to the distal criterion in the probabilistic functionalism theory) and is recovered empirically from past organizational decisions. Figure 1 shows the resulting model, which we call the Contextualized Alignment Lens Model (CALM).

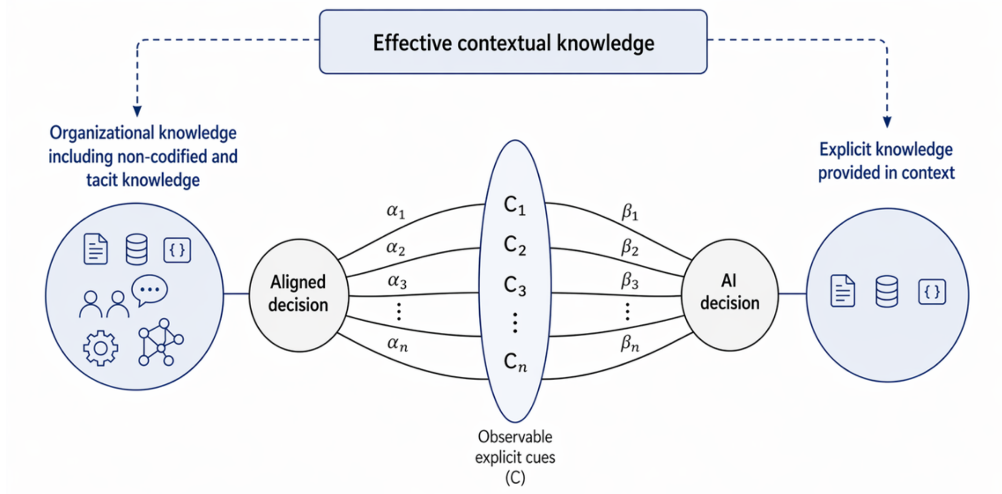

Figure 1. The Contextualized Alignment Lens Model (CALM).

Model Description

CALM treats the organizational decision-maker and the LLM as two judges acting on a shared cue set $C = \{C_1, ..., C_n\}$, where each cue is an observable feature of a decision case. The left side of the lens represents the aligned decision policy, that is, the cue-weighting policy that the organization applies in practice and that the LLM should reproduce to be contextually aligned. We recover this policy empirically from past organizational decisions, on the assumption that decisions made under the organization's full knowledge base reveal the policy it endorses. The aligned policy is therefore grounded in the full "stack" of organizational knowledge including codified records, non-codified practice, and tacit know-how. The right side represents the AI decision policy. Because the LLM accesses only the textual context provided to it, this policy is grounded in a strict subset of organizational knowledge, namely the part that is explicit (available in codified form or made explicit through knowledge externalization) and supplied in the prompt or made available for retrieval. Both policies are mediated by the same observable cues, which makes them directly comparable.

We estimate each policy by regressing decisions on cues. On the organization side, a regression of past decisions on cues yields a coefficient vector $\alpha = (\alpha_1, ..., \alpha_n)$ that captures how the organization weights cues in practice. On the LLM side, an analogous regression of model decisions on the same cues yields $\beta = (\beta_1, ..., \beta_n)$. Because α is fitted on past organizational outcomes and β on outputs of the contextualized model, α reflects the organization's effective use of cues across the full knowledge base, while β reflects the model's use of cues given only the explicit context.

We refer to the agreement between α and β as *process alignment* or *policy alignment* and measure it with a similarity score $s(\alpha, \beta)$ bounded on the unit interval, with higher values indicating closer agreement on cue-specific weights. We adopt cosine similarity as the primary metric because it is symmetric, scale-invariant, and bounded, which makes it interpretable across models and conditions. As a robustness check, we rely on Pearson correlation of the coefficient vectors. We assess statistical significance with a bootstrap permutation test that shuffles the cue–coefficient correspondence in β, which tests whether agreement reflects shared cue-specific weighting rather than coincidental magnitude similarity.

The ideal state in CALM corresponds to what Brunswik (1956) probabilistic functionalism calls *achievement*. When full achievement is reached, the LLM's cue-weighting policy matches the organizational policy and $s(\alpha, \beta)$ approaches one. Two further constructs from that theory have natural counterparts in CALM. *Cue validity*, the predictive relation between a cue and the distal criterion (i.e. the aligned decision in CALM), corresponds to the α coefficient of a cue, which captures how strongly the organization relies on that cue in practice. *Cue utilization*, the weight a judge places on a cue, corresponds to β. A third construct, *vicarious functioning*, describes a judge's substitution among cues when the cue environment changes, and predicts that a well-aligned LLM should reweight cues in expected ways when the available cue set is perturbed. We do not test for vicarious functioning in our demonstration, but see it as a promising direction for further development.

Diagnostic Information

CALM produces four classes of diagnostic information.

First, it localizes misalignment to specific cues. The cue-level differences between α and β indicate which features the model under-weights, over-weights, or weights with the wrong sign. This level of detail supports targeted interventions: A developer can revise the prompt, retrieve additional documents, or restructure tool descriptions to address the specific cues on which the policies diverge. Then, CALM can be applied once again to re-estimate β and measure whether the intervention improved alignment.

Second, CALM exposes unwanted reliance on cues that the organization would reject. A cue with a substantial β coefficient and a negligible or normatively prohibited α coefficient signals a process-level bias, even when outputs happen to agree with the organization on aggregate. In hiring decisions, for instance, a significantly non-zero β on a gender cue could surface an unwanted bias. CALM thus complements existing approaches to explainable AI (Guidotti et al., 2018) by making the implicit decision policy of a contextualized LLM auditable at the cue level.

Third, CALM provides a bounded estimate of how much tacit or unarticulated knowledge a decision draws on. While it cannot fully distinguish among the components (tacit knowledge, incomplete articulation, model-side limitations) of the residual gap, it can provide an upper bound estimate through iterative application after knowledge externalization interventions: Once the alignment improvements from externalization interventions plateau but the cue set explains a substantial share of decision variance on the organization side, the residual gap between α and β is an upper bound estimate on the influence of tacit knowledge on the decision. It is an imperfect estimate because the residual also absorbs articulation failures, model-side limitations such as imperfect instruction following, and limitations from the linear-assumption of the regression. Nevertheless, a small residual under these conditions can suggest that the specific decision is more prone to automation, while a large residual can suggest that human involvement remains warranted to include the necessary tacit knowledge to achieve an aligned decision (e.g. through a human-in-the-loop setup).

Fourth, CALM supports comparison across models and across interventions. Because α is fixed for a given organizational setting, alternative LLMs and alternative context configurations can be evaluated against the same target. This produces a model-selection criterion that is distinct from accuracy, namely the degree to which a model's process matches the organization's process under a given alignment intervention.

Prerequisites

CALM can be applied under four conditions. The first is the availability of past decision data with sufficient coverage of the case features that the decision is presumed to depend on. Without past decisions, α cannot be estimated and the comparison is undefined. The second is that the observable cue set captures a meaningful share of the variance in past decisions. We operationalize this through the predictive performance of the α-side regression, for example via the area under the receiver operating characteristics curve (AUC) in the binary case. A low value indicates that the cue set has limited explanatory power, which makes the comparison of weighting policies uninformative regardless of how β is estimated. The third is that a linear cue model is an adequate approximation of the decision policy. Where for example strong cue interactions or thresholding effects dominate, the α and β estimates are still defined but their comparison may understate the true policy similarity. This condition can be addressed by extending CALM to non-linear cue functions in the future. Finally, CALM is most informative when the model-side decisions exhibit minimal task competence or policy stability. If LLM decisions are near chance or weakly related to the target decision, β remains mathematically estimable and might still provide useful diagnostic information for designing knowledge interventions, but loses the rest of the standalone interpretability as suggested above.

CALM Demonstration

We demonstrate CALM on European Court of Human Rights (ECtHR) decisions on Article 6 (right to a fair trial). We use the collection released by Chalkidis et al. (2021), which – among other features – contains textual case descriptions paired with the court's binary decisions on whether the article was violated. From this collection we sampled 1,000 cases balanced by outcome. ECtHR Article 6 fits CALM well: a decision by a stable institution that reasons systematically across many cases, standardized facts statements contain the textual material the court itself works with, and ground-truth outcomes support recovery of the organizational policy.

Cue Codebook and Coding

Operationalizing CALM requires a cue set that captures a meaningful share of variance in past organizational decisions. For ECtHR Article 6 we constructed a codebook of 45 binary cues describing observable case features, derived from the jurisprudential literature on Article 6 through three independent AI deep-research tools whose reports we consolidated. Each cue is defined as 1 if the feature is explicitly present in the case facts and 0 otherwise. We coded all 1,000 cases automatically using GPT-5.4-mini with the codebook supplied as a single prompt requesting judgements on all 45 cues per case. To select an LLM coder we conducted a pretest on the first 100 cases comparing GPT-5.2, GPT-5.4-mini, GPT-5.4-nano, and Claude Haiku 4.5. GPT-5.4-mini reached Cohen's $\kappa = 0.738$ with the larger reference model GPT-5.2 (90.7% agreement), the strongest agreement among the smaller candidates, and was selected for the full coding pass. We validated the codebook by evaluating its explanatory power. A the cross-validated AUC for the α-side regression yielded 0.715. This is sufficient to conclude that a sufficient amount of variance in the court's decisions is captured by the coded cues for CALM to be applicable.

Models, Conditions, and Decision Procedure

We evaluated ten LLMs selected to span sizes, providers, and geographical origin: GPT-5.4, GPT-5.4-mini, and GPT-5.4-nano (OpenAI), Claude Haiku 4.5 (Anthropic), Gemma 4 31B (Google), Grok 4.1 Fast (X.ai), Mistral Large (Mistral AI), DeepSeek-v3.2 (DeepSeek), Minimax M2.7 (Minimax), and Qwen 3.6 Plus (Alibaba). Models were accessed through the OpenAI API and OpenRouter API using each provider's default sampling settings. Where a model exposed a configurable reasoning effort, we set it to medium.

Each model decided each case under three conditions. The system prompt was identical across conditions and instructed the model to act as a legal assessment assistant for ECtHR cases, to use only the provided case facts and Article 6 text, and to make a forced binary prediction. The user prompt contained the case identifier, title, judgment date, the verbatim text of Article 6, the case facts, and a forced-choice instruction to begin the response with "V" (violated) or "N" (not violated) followed by a brief reasoning. The three conditions differ only in whether and how externalized cue-weighting information is appended to the user prompt before the case facts.

Baseline. No externalized cue-weighting information is provided. The model decides on the basis of the Article 6 text and the case facts alone. This condition measures the LLM's default decision policy when the only contextual knowledge is the case description and the article 6 text on top of what the model has acquired during training.

Court-externalized. A summary of the court's most heavily weighted cues is appended to the prompt. We selected the top fifteen cues by absolute α coefficient from the organization-side ridge regression ("aligned decision") and described them qualitatively in plain language as strong, moderate, or weak indicators in the direction implied by the sign of their coefficients.

Introspective-externalized. A model-specific summary of where the LLM's baseline policy differs most from the court's is appended to the user prompt. For each model, we identified the fifteen cues with the largest absolute difference between the model's baseline β and the court's α, and provided cue-specific adjustment guidance describing the direction and magnitude of the required shift. This condition tests whether direct correction of the model's observed policy gap improves alignment, and it differs from the court-externalized condition by targeting the model's specific divergence rather than the organization's general policy.

Both externalization conditions follow one hypothesis: By externalizing implicit organizational knowledge embedded in past decisions through language instructions, we narrow the gap between what the LLM has access to about the organization's policy and what the organization itself draws on when deciding. This shift in the LLM's epistemic position should move its cue-utilization policy toward the organization's. We therefore expect each intervention to increase $s(\alpha, \beta)$ relative to baseline, an expectation tested with one-sided paired bootstrap.

Estimation and Significance Testing

We recover the cue-utilization policy on each side of the lens by ridge logistic regression of binary decisions on the 45 binary cues. The organization-side α-policy is estimated from the 1,000 ECtHR decisions using the coded cues. The LLM-side β-policy is estimated separately from a specific model's 1,000 decisions under one of the three conditions, using the same cue values. Ridge regularization is preferred over unpenalized logistic regression because the penalty stabilizes coefficient estimates under collinear cues, which is the relevant property for comparing weighting policies across regressions fit on different decision distributions.

We test whether each externalization intervention shifts alignment relative to baseline using a paired bootstrap procedure on the difference $s(\alpha, \beta_{intervention}) - s(\alpha, \beta_{baseline})$, with 500 paired resamples and one-sided p-values reported in the direction of expected improvement.

Results

We report the ECtHR demonstration findings in three subsections: (1) baseline process alignment across models, (2) effects of externalization interventions, and (3) the case-level relationship between process and outcome.

Baseline Process Alignment

Baseline process alignment varies widely across the ten models. Cosine similarity between the court's α and the model's β ranges from -0.211 (GPT-5.4-nano) to +0.844 (GPT-5.4-mini). Four models reach baseline alignment indistinguishable from zero, with 95% bootstrap confidence intervals that span zero or fall below it: Mistral Large, DeepSeek v3.2, Claude Haiku 4.5, and GPT-5.4-nano. The remaining six models reach positive baseline alignment, and three of them (GPT-5.4, GPT-5.4-mini, Grok 4.1 Fast) cluster above 0.82. Table 1 reports the baseline cosine similarity, accuracy, Cohen's κ, and AUC for each model.

Model	Cosine α–β	Accuracy	Cohen's κ	AUC
GPT-5.4-mini	0.844	0.725	0.450	0.770
Grok 4.1 Fast	0.842	0.710	0.420	0.751
GPT-5.4	0.824	0.721	0.442	0.755
Qwen 3.6 Plus	0.750	0.671	0.342	0.751
Gemma 4 (31B)	0.710	0.637	0.274	0.781
Minimax M2.7	0.592	0.561	0.122	0.654
Mistral Large	0.083	0.550	0.100	0.725
DeepSeek v3.2	0.062	0.513	0.026	0.515
Claude Haiku 4.5	-0.057	0.494	-0.012	0.797
GPT-5.4-nano	-0.211	0.462	-0.076	0.719

Table 1. Baseline metrics for each model.

This spread does not track general-purpose model capability or pricing. Claude Haiku 4.5 (Anthropic) is approximately five times more expensive per token than Grok 4.1 Fast (xAI). On baseline contextualized alignment in our setting, the ordering reverses: Claude Haiku 4.5 reaches a cosine of -0.057, while Grok 4.1

Fast reaches +0.842. This pattern also applies to other model pairs. Consequently, pricing and general benchmark performances are weak predictors of how closely a model's cue-utilization policy will match a specific organization's policy at deployment.

In general, we find that process and outcome alignment correlate strongly across all 30 model–condition combinations. Pearson r between cosine similarity and outcome accuracy is 0.85 ($p < 0.001$), and Spearman ρ is 0.94 ($p < 0.001$). Figure 2 shows the corresponding scatter plot. The strength of this correlation supports the claim that outcome accuracy in this setting is largely a function of the policy a model applies, not of its general capability.

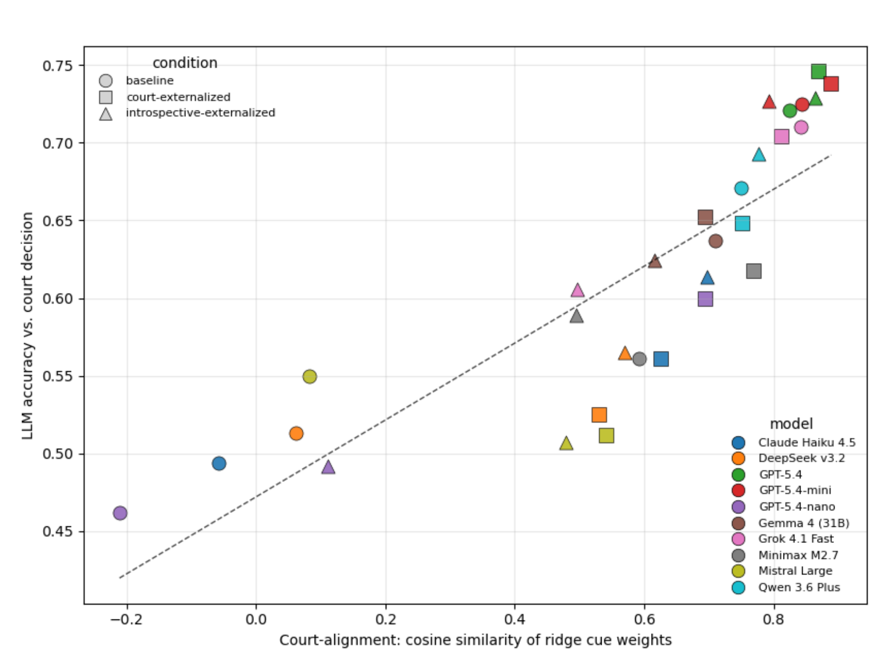

Figure 2. Correlation between policy alignment and accuracy.

The α-side ridge regression of court decisions on the 45 binary cues achieves a cross-validated accuracy of 68.5%. Three baseline LLMs already exceed this score: GPT-5.4 (72.1%), GPT-5.4-mini (72.5%), and Grok 4.1 Fast (71.0%). These models therefore use information in the case facts that the codebook does not capture or the linearity assumption of the regression loses. We return to this point in the following section when discussing how externalization interventions transfer to outcome accuracy.

Effects of Externalization Interventions

We test the effects of the two externalization interventions. Both interventions are deliberately simple in form. They translate CALM's cue-level diagnostic information directly into prompt content. The court-externalized condition surfaces the cues the organization weights most heavily, and the introspective condition surfaces the cues on which the model's β diverges most from the court's α. Better intervention designs are likely possible, but the interventions we test isolate the contribution of CALM's diagnostic information itself: any improvement we observe is attributable to the cue-level guidance the method produces. Figure 3 shows the per-model alignment under all three conditions.

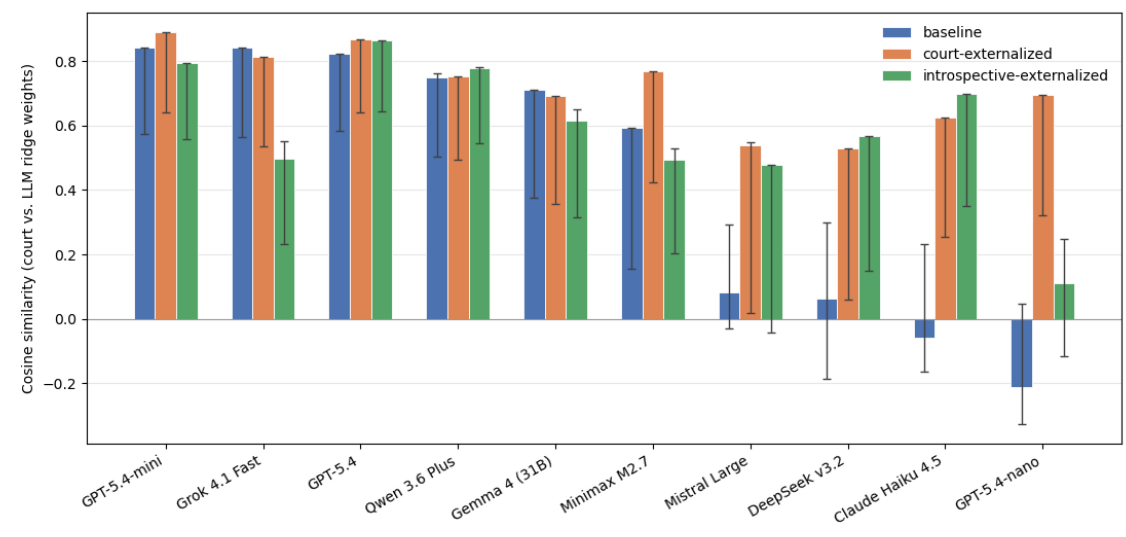

Figure 3. Decision policy alignment across models and conditions.

Court-externalization moves the cosine point estimate in the predicted direction for 8 of 10 models, and produces a statistically significant improvement (one-sided $p < 0.05$) for three of them: Claude Haiku 4.5, GPT-5.4-nano, and Minimax M2.7. Introspective-externalization moves the point estimate in the predicted direction for 6 of 10 models and produces a significant improvement for three: Claude Haiku 4.5, DeepSeek v3.2, and GPT-5.4-nano. The directional rate is consistent with our hypothesis that supplying organizational knowledge through the prompt improves process alignment, but the inferential strength of the effect varies sharply across models.

The effect size depends on baseline alignment. Models with weak or negative baseline alignment improve by large margins under court-externalization: GPT-5.4-nano shifts from -0.211 to +0.695 (Δ = +0.906), and Claude Haiku 4.5 shifts from -0.057 to +0.625 (Δ = +0.682). Models with already-high baseline alignment shift much less. GPT-5.4 moves from 0.824 to 0.869 (Δ = +0.045), and GPT-5.4-mini moves from 0.844 to 0.889 (Δ = +0.045). The pattern is consistent with a ceiling effect: when a model's policy already matches the organization's well, externalization has little room to improve it.

Nevertheless, for some of the high scoring baseline models, the externalization interventions still improved process alignment and outcome accuracy. GPT-5.4 and GPT-5.4-mini improve their process alignment and outcome accuracy in at least one of the two externalization interventions, although their baseline accuracy was already higher than the aligned ridge model. The CALM diagnostic information is therefore useful even for models that perform well at baseline.

These results have implications for model selection. Two model properties matter independently: baseline contextual fit and intervention responsiveness. They are partly orthogonal in our data: Claude Haiku 4.5 has weak baseline fit but high responsiveness, GPT-5.4 and GPT-5.4-mini have strong baseline fit and small but stable responsiveness, and Grok 4.1 Fast has strong baseline fit but regresses under introspective intervention. Neither axis tracks general benchmark performance or pricing tier.

A finer-grained analysis on correct and incorrect decisions shows that process and outcome alignment are connected not only across models but also within models. Across all ten models at baseline, mean cosine similarity between α and β computed on the subset of cases where the model agrees with the court is 0.811. On the subset where the model disagrees with the court, the same mean is -0.303. Cases where the LLM reaches the wrong outcome are also cases where its cue-utilization policy diverges most from the court's.

Externalization interventions narrow this gap, primarily by improving alignment on incorrect cases. Court-externalization raises the mean cosine on incorrect cases from -0.303 to +0.033 (a shift of +0.336 averaged across models), while raising the mean on correct cases only from 0.811 to 0.876 (a shift of +0.065).

Introspective-externalization produces a smaller version of the same pattern. Interventions therefore reduce the difference between the policy a model applies on cases it gets right and the policy it applies on cases it gets wrong.

The cue-shift plots in Figure 4 show that the improvements are produced through the mechanism the externalization interventions based on CALM targeted. Under court-externalization (left in figure), most prompt-targeted cues move on average in the direction the diagnostic prescribed for most models. Under introspective-externalization (right in figure), the same is true on average but the heterogeneity across models is larger. Some models adjust the targeted cues precisely, others overcorrect on cues the prompt singles out for emphasis, and one model (Grok 4.1 Fast) fails entirely.

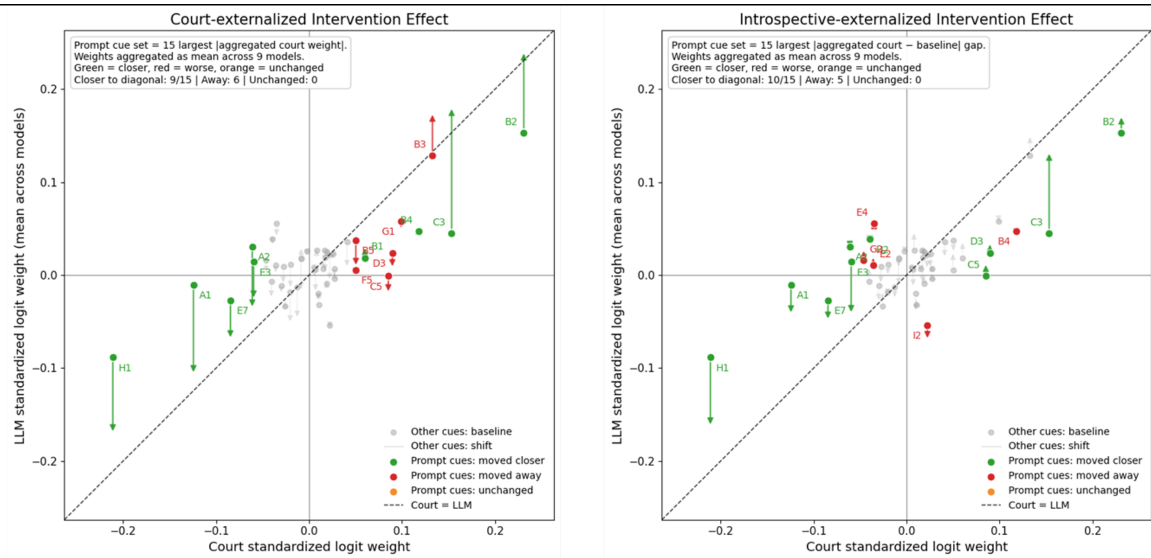

Figure 4. Cue weighting effect for the court-externalized and the introspective-externalized knowledge interventions.

The Grok 4.1 Fast result illustrates this heterogeneity at its extreme. The model regresses sharply under introspective-externalization, dropping from a baseline cosine of 0.842 to 0.496 (one-sided p = 0.002 in the worsening direction). The Figure 4 diagnostic for this models shows that this was mostly caused by an overadjustment of the mentioned cues and many large movements in cue-weightings of cues that were not mentioned in the intervention prompt. Both are typical failure modes that we observed in different strengths across models. Consequently, CALM's diagnostic information is a necessary input for principled intervention design but does not on its own determine the outcome of the intervention: how the model integrates the diagnostic information matters and the quality of the codebook (e.g. avoiding correlation in between cues) matter as well. CALM can help in surfacing this information.

A robustness check with Pearson correlation instead of cosine similarity confirmed our results.

Discussion

Theoretical Implications

We make two contributions to the IS literature on the development and design of information systems that incorporate LLM decision-making.

The first is conceptual. We argue that contextualized alignment is needed because LLMs do not automatically occupy the same epistemic position as the organizations in which they are deployed. Organizational decisions draw on a stack of knowledge that includes codified records, explicit rules, non-codified routines, situated norms, and tacit know-how, only some of which can be transferred to an LLM through textual context (Lebovitz et al., 2021; Pakarinen & Huising, 2023; Polanyi, 1962, 1966). This knowledge gap explains why general AI alignment, which focuses on matching broad human intentions and values (Gabriel, 2020; Ji et al., 2025; Russell, 2019), is necessary but insufficient for organizational deployment. Contextualized alignment builds on this prerequisite as an application-level alignment problem. Once an LLM's behavior is aligned in a broadly ethical and generally acceptable way, it still needs to fit the situated decision context in which it is used. The relevant question for applications is whether the model has access to enough of the organization's decision-relevant knowledge to reproduce the organization's decision policy in a specific sociotechnical context. Contextualized alignment therefore emphasizes the effects caused by differences in the knowledge structure of AI and organizational decision-makers. In this setting, we interpret in-context learning capacity of LLM as a form of organizational knowledge externalization in the Nonaka (1994) sense. The prompt is a medium through which implicit organizational knowledge is converted to explicit form and transferred to the LLM. Our proposed method provides a way to measure the success of the transfer.

The second contribution is methodological. We build on the Brunswik lens model, widely used for the analysis of expert judgment (Brunswik, 1956; Hammond et al., 1964; Karelaia & Hogarth, 2008), to compare an organization's decision policy with that of an LLM put in the same situation. CALM operationalizes process alignment through the cosine similarity of cue weight vectors recovered by regression on each side of the lens. This move turns the abstract sociotechnical alignment question (whether the AI subsystem is aligned to the values of the sociotechnical system it operates in) into an empirical comparison between α and β.

Practical Implications

The findings carry two implications for organizations developing AI systems that rely on LLM decisions.

First, general-purpose benchmarks and pricing tier are weak predictors of contextualized fit. Models in our sample with similar performance on public benchmarks differ drastically on cosine similarity in their baseline alignment with the court's policy. Two model properties matter independently for deployment: baseline contextual fit, that is, how closely the off-the-shelf model's policy matches the organization's, and intervention responsiveness, that is, how reliably the model shifts toward the target policy when externalized knowledge is supplied. CALM provides a way to evaluate both axes empirically before deployment, in contrast to selection criteria that rely on benchmark accuracy alone.

Second, the residual α–β gap supports a principled approach to automation potential and human-in-the-loop placement. Under the conditions that the alignment improvement effect of externalization interventions plateaus and the cue set explains a substantial share of variance, the residual gap places an upper bound on the contribution of tacit knowledge to a given decision. Where the bound is small, the case for delegating the decision to an LLM is stronger. Where the bound is large, human involvement is warranted to bring the tacit knowledge that the prompt cannot transmit. This framing is one input among several into automation decisions and does not displace deliberation on accountability, fairness, and oversight, but it provides a quantifiable signal that previous approaches to automation readiness lack. The pattern is consistent with empirical findings on AI overruling in professional practice (Schmitt et al., 2025), where practitioners systematically overrule AI advice on cases that draw on knowledge the AI does not have.

Limitations

CALM relies on the availability of past decision data with sufficient coverage of the case features the decision is presumed to depend on. Where past decisions are not available, recoverable, or representative of the policy the organization currently endorses, the α-policy cannot be estimated and CALM does not apply. This limits the immediate scope of the method to settings with documented decision histories.

Our operationalization recovers α from past organizational decisions and treats it as a reasonable approximation of an informed decision-maker's policy in stable institutional settings. The normative definition of process alignment allows for divergence between past decisions and reflective endorsement, particularly where past decisions encode biases the organization itself would reject. CALM does not by itself resolve this gap. The bias-detection affordance of CALM partly addresses it by surfacing reliance on cues that the organization rejects, but the alignment target remains an empirical artifact. In practice, this requires that researchers and practitioners applying CALM treat dataset selection as a normative decision: scoping the past-decision set to a period, unit, or process the organization currently endorses, and excluding decisions with known biases that have since been corrected.

We focus on in-context learning as the alignment intervention because it is the most accessible to deploying organizations and does not require modifying model weights. Alignment through fine-tuning or pre-training on organization-specific data is also feasible but outside our scope. It may close portions of the residual gap that prompting cannot, but fundamental limitations will remain (Lebovitz et al., 2021).

Finally, there are some limitations in the modeling capacity of CALM. These include that the linear-additive cue model may understate policy similarity where relationships are non-linear and that the codebook may not capture all decision-relevant cues.

Future Research

Three directions are worth highlighting.

First, the method should be tested in settings with greater inter-organizational variation in decision policy. ECtHR Article 6 is a stable single-institution setting that supports a clean demonstration but offers limited variation in the policy CALM is comparing against. Insurance claim adjudication, hiring decisions, and medical triage are settings in which different organizations might apply different policies to similar inputs. CALM should be able to surface those differences across organizations and give a tool to adjust LLM decisions to these specific settings.

Second, the empirical α policy should be compared with the policy LLMs *report* themselves to apply. Work on chain-of-thought faithfulness (Lanham et al., 2023; Turpin et al., 2023) and shortcut learning (Geirhos et al., 2020) shows that stated reasoning can diverge from the actual cue-utilization policy. Asking each model to declare its cue weights and comparing the declared weights to the empirical β estimated by CALM would measure the size of this introspection gap and connect process alignment to interpretability research.

Third, the responsiveness profile observed in our results suggests a new dimension for LLM evaluation. Models in our sample differ along two partly orthogonal axes: baseline contextual fit and responsiveness to externalization interventions. Neither axis tracks general benchmark performance. A benchmark that operationalizes both axes across organizational decision settings would complement existing capability and safety benchmarks with a contextual-fit measure. Vicarious functioning, another construct from probabilistic functionalism, is one promising mechanism for stress-testing models on this dimension by perturbing the cue set and observing whether the LLM substitutes among cues in the way the organization does. We also see value in extending CALM with cue-type analysis to identify which classes of features are most likely to encode tacit organizational knowledge that prompting struggles to convey.

Conclusion

LLMs are increasingly used to take consequential decisions in organizational settings, yet much of the knowledge that grounds those decisions resists complete transfer through textual context. We introduced contextualized alignment to name this gap and proposed the Contextualized Alignment Lens Model (CALM) to measure it. Building on the Brunswik lens, CALM compares the cue-weighting policy revealed in past organizational decisions with the policy a contextualized LLM applies to the same cases. The comparison localizes misalignment to specific cues, surfaces reliance on cues the organization would reject, supports targeted prompt-level interventions, and yields a bounded estimate of the contribution of tacit knowledge to a given decision.

The demonstration on 30,000 LLM decisions across ten models and three prompting conditions on ECtHR Article 6 shows that baseline alignment varies widely in ways that general benchmarks and pricing do not predict, that prompt-based knowledge externalization can move cue utilization toward the organizational policy with the largest effects on the most misaligned starting points, and that process and outcome alignment correlate strongly across model–condition combinations. Taken together, these findings indicate that situated decision quality depends as much on policy fit as on general model capability. CALM operationalizes contextualized alignment and gives IS researchers and practitioners a concrete instrument for evaluating, comparing, and improving LLM decision-making in the sociotechnical systems where it is deployed.

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